

EXPERIMENT 3

Aim

To apply Decision Tree and Random Forest algorithms on a real-world dataset to perform classification and compare their performance in detecting fraudulent transactions.

Problem Statement

Credit card fraud detection is a critical problem where transactions must be classified as:

- Fraudulent (1)
- Non-Fraudulent (0)

The dataset is highly imbalanced, making it challenging to detect fraud accurately. The objective is to:

- Build a Decision Tree model for classification
- Build a Random Forest model to improve accuracy and reduce overfitting
- Compare both models based on performance metrics

Dataset Link

- Dataset Name: Credit Card Fraud Detection Dataset
- Source: Kaggle
- Link: <https://www.kaggle.com/code/abdelazizelserty/credit-card-fraud-detection-dataset>

Methodology

1. **Data Collection**
 - Loaded the dataset (`creditcard_2023.csv`)
2. **Data Preprocessing**
 - Separated features (X) and target variable (y)
 - Removed missing values from the target column
 - Ensured data consistency by aligning X and y
3. **Data Splitting**
 - Split dataset into training and testing sets (80:20 ratio)
 - Used stratified sampling to handle class imbalance
4. **Decision Tree Model**
 - Trained Decision Tree Classifier
 - Used default parameters with random state
5. **Random Forest Model**
 - Trained Random Forest Classifier with:
 - 100 trees (`n_estimators=100`)

- Parallel processing (n_jobs = -1)

6. Model Evaluation

- Accuracy Score
- Confusion Matrix
- Classification Report (Precision, Recall, F1-score)

7. Model Comparison

- Compared accuracy of Decision Tree and Random Forest

Result

Decision Tree Results

- Achieved good classification performance
- Easy to interpret and visualize
- Tends to overfit on training data, especially with imbalanced datasets

Random Forest Results

- Achieved higher accuracy compared to Decision Tree
- Reduced overfitting using ensemble learning
- More robust and stable for fraud detection

Output:

```
----- Decision Tree Results -----
Accuracy: 0.9985679507375054
Confusion Matrix:
[[6956   6]
 [  4  17]]
Classification Report:
              precision    recall  f1-score   support

    0.0         1.00      1.00      1.00     6962
    1.0         0.74      0.81      0.77        21

 accuracy          0.99
 macro avg         0.87      0.90      0.89
 weighted avg         0.99      0.90      0.89
```

```

----- Random Forest Results -----
Accuracy: 0.9994271802950021
Confusion Matrix:
[[6959   3]
 [   1  20]]
Classification Report:

```

	precision	recall	f1-score	support
0.0	1.00	1.00	1.00	6962
1.0	0.87	0.95	0.91	21
accuracy			1.00	6983
macro avg	0.93	0.98	0.95	6983
weighted avg	1.00	1.00	1.00	6983

```

Accuracy Comparison
Decision Tree Accuracy: 0.9985679507375054
Random Forest Accuracy: 0.9994271802950021

```

Conclusion:

In this experiment, Decision Tree and Random Forest classifiers were implemented on the Credit Card Fraud Detection dataset. While the Decision Tree model was simple and interpretable, it showed signs of overfitting. The Random Forest classifier performed better due to ensemble learning and provided more accurate and reliable results. Hence, Random Forest is more suitable for fraud detection tasks.